Project Requirements & Collaboration Outline

Drone Backend & Workflow for Regenerative Agriculture — Missouri S&T Multirotor Design Team

Document Version:	Version 1.1	**Date:**	August 2026
Primary Lead:	Allseeing Research Services	**Development Lead:**	Matthew Wasser (Missouri S&T Multirotor)

Revision Summary (Version 1.1)

Updated per feedback from Matthew Wasser (Missouri S&T SWEB/Multirotor Lead):

- Re-sequenced roadmap milestones: Photogrammetry ingestion, local UI, and browser-based 3D model display are now Milestone 1 to allow immediate testing without hardware constraints.
- Expanded flight planner specs to evaluate multi-platform export support (ArduPilot/MAVLink mission formats alongside DJI WPML).
- Acknowledged hardware testing constraints (S&T team currently possesses a Mavic 2 Pro; field validation with Mavic 4 Pro handled by Allseeing).

1. Project Purpose & Scope

The goal of this collaboration is to design, build, document, and deliver a practical, user-friendly drone software workflow tailored for regenerative agriculture research and farm management. The software operates in two distinct modes:

- **Standalone Mode:** A simple UI/CLI that users can execute on their own local hardware or servers.
- **Backend / API Mode:** A RESTful backend service that integrates seamlessly with the **SoilDaddy** farm workspace and feeds geospatial photogrammetry products into the **MoPine** MBSE digital twin.

Key Operational Boundaries:

- Missouri S&T is responsible for building and delivering runnable, documented software. Long-term compute hosting or cloud infrastructure will *not* be provided by Missouri S&T post-delivery.
- Allseeing Research Services is responsible for all field operations (flying drone missions), capturing imagery, and uploading image sets into the shared processing cache.
- Missouri S&T will focus strictly on software execution and photogrammetry processing—data analysis and agronomic interpretation are handled by Allseeing and research partners.

2. Responsibility Matrix

Partner / Entity	Assigned Responsibilities
Allseeing Research Services	Overall coordination, system requirements, SoilDaddy integration, drone flight capture (Mavic 4 Pro), uploading imagery into cache/system, and agronomic analysis.
Missouri S&T Multirotor Team	Design, build, document, and deliver runnable software (processing pipeline + 3D web viewer + flight planning abstraction) supporting standalone and backend API modes.
SoilDaddy	Front-end user interface, parcel/paddock selection, map displays, and API calls to the drone backend.
Elm Spring Farms (Matt Adler)	Primary pilot/research site (Hughesville, MO), mission support, and domain feedback.
MoPine (Todd Sparks)	MBSE digital twin platform consuming processed photogrammetry and spatial outputs.
Lincoln University	Academic research partner and agricultural domain validation.

3. Hardware & Protocol Baselines

- **Primary Production Platform:** DJI Mavic 4 Pro (25 MP baseline, 100 MP stretch option) — flown and validated in field operations by Allseeing.
- **S&T Development Platform:** DJI Mavic 2 Pro (used locally by S&T team for development/testing context).
- **Flight Controller & Protocol Flexibility:** Architecture will research and build towards flight plan abstraction supporting both DJI formats (WPML XML) and open-source flight controllers running **ArduPilot** / **MAVLink** (e.g., QGroundControl `.plan` format).
- **Format Flexibility:** Ingestion pipeline accepts standard geotagged EXIF (JPEG/TIFF) image sets from any aerial platform.

4. Required Technical Capabilities

A. Image Processing & 3D Web Visualization (Priority Baseline)

- **Data Ingestion & Pipeline:** Process geotagged imagery into core photogrammetry outputs using OpenDroneMap / WebODM core.
- **In-Browser 3D Rendering:** Integrated web viewer (e.g., Three.js, CesiumJS, or Potree) allowing users to inspect textured 3D meshes (`.gltf`/`.obj`) and dense point clouds directly in the browser.
- **Core Required Outputs:**
  - Georeferenced RGB Orthomosaic (Cloud-Optimized GeoTIFF format preferred).
  - Digital Surface Model (DSM).
  - Processing quality / spatial accuracy report.
  - Standardized JSON metadata package (capturing bounds, GSD, execution parameters).

- **Strongly Desired Outputs:**
  - Canopy Height Model (CHM) derived via DSM/DTM subtraction.
  - RGB Vegetation Indices: VARI, TGI, ExG, and GRVI raster layers.
  - Fractional Canopy / Vegetation Cover percentage estimation.
  - Multi-temporal change detection support.

B. Multi-Platform Flight Planning Subsystem

- Accept Area of Interest (AOI) polygon inputs via GeoJSON or shape interface (standalone UI or SoilDaddy API).
- Configure parameters: target Ground Sampling Distance (GSD) / altitude, front/side overlap percentages, and sensor profile.
- Generate optimal lawnmower/grid flight paths and estimate image count, flight time, and battery needs.
- **Multi-Format Exporter Architecture:** Research and implement export modules for:
  - **ArduPilot** / **MAVLink:** Standard open waypoint format (`.plan` / Waypoint file).
  - **DJI WPML:** XML schema packaged in `.kmz` archives for DJI Fly / Mavic 4 Pro compatibility.
- Support automatic mission partitioning/splitting for large AOIs exceeding single-battery thresholds.

C. Architecture & Integration

- **Standalone Interface:** Clean GUI or Web UI (e.g., Streamlit/FastAPI) requiring no engineering expertise.
- **Backend REST API:** Endpoints for AOI submission, flight plan generation, job queuing, status polling, and asset retrieval.

5. Revised Milestone Roadmap

1. **Milestone 1 (Immediate Focus):** Simple standalone interface & photogrammetry pipeline taking in geotagged image sets to generate and display interactive 3D renders (mesh/point cloud) directly in the browser.
2. **Milestone 2:** Core 2D deliverables (Cloud-Optimized Orthomosaic + DSM + Quality Report + JSON metadata sidecar) runnable locally by Allseeing.
3. **Milestone 3:** RESTful Backend API integration for SoilDaddy orchestration and status tracking.
4. **Milestone 4:** Flight Planning Subsystem — Grid generation, multi-vehicle abstraction research, ArduPilot (`.plan`), and DJI WPML (`.kmz`) exporters.
5. **Milestone 5:** Advanced Photogrammetry Products (CHM, RGB vegetation indices, change detection) and end-to-end field testing with Allseeing imagery.

6. Team Review & Incorporation Notes

A. Technical Feasibility & Scope Adjustments (Incorporated from M. Wasser):

> Roadmap updated to start with image processing & web-based 3D visualization. Research spike added to evaluate ArduPilot vs. DJI flight plan exports to maintain cross-platform flexibility.

B. Preferred Communication & Code Delivery Structure:

[Missouri S&T Team: Insert repository links, meeting cadence, or preferred Discord/Slack channels]

C. Estimated Milestone Schedule:

[Missouri S&T Team: Insert target dates for Milestones 1 through 5]